



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

Nuclear Starburst Disks

ACTIVE GALACTIC NUCLEI

Fritz Ali Agildere

April 29, 2026

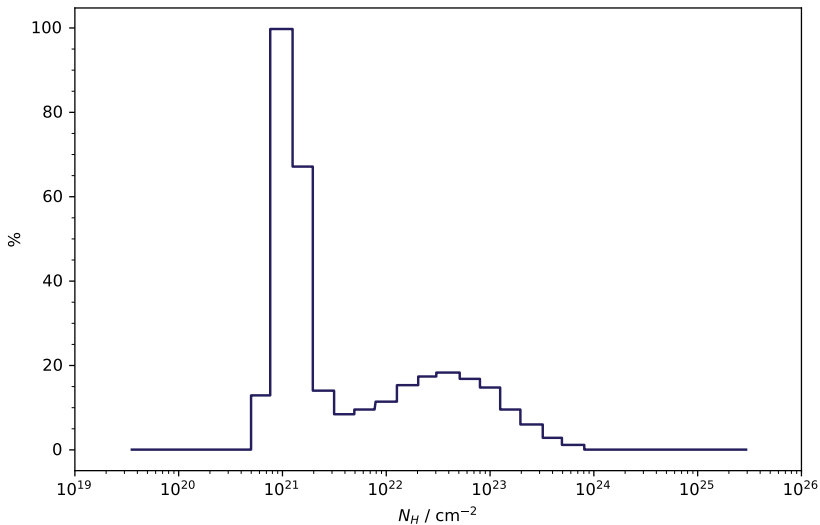
Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Fuel Region / Galactic Disk Kinematic Decoupling
- Spatial SF & AGN Luminosity Correlation
 - Galactic Scales $R \sim 10 \text{ kpc}$ Weak
 - Compact Nucleus $R \sim 100 \text{ pc}$ Strong
- Vertical Gas & Dust Expansion Intercepts Sightlines
- Dynamically Evolving Obscuring Starforming Structure

Structure

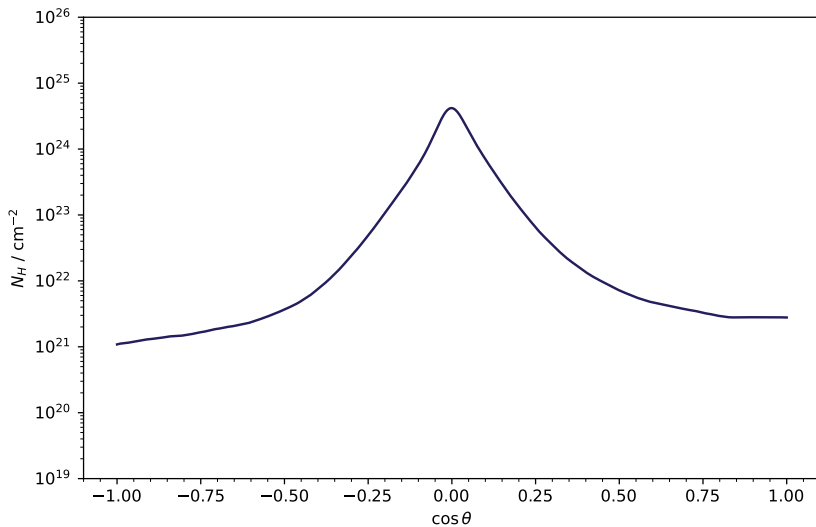
- Radiatively Supported Large Scale Heights (Star Formation)
 - Intense Stellar Radiation Pressure
 - Supernovae Kinetic Feedback
- Distinct Peripheral $R > 10 \text{ pc}$ Regions (Compton Thin)
- Highly Localized $R < 1 \text{ pc}$ Obscuration (Compton Thick)
- Embedded Stellar Remnant Thermal Effects
- Statistically Consistent Viewing Angle Sampled Distributions

Column Density



Modeling

Angular Obscuring



Outlook

- Observational Static Torus / Puffy Compact Starburst Disk Separation
- Evolutionary Stage / Static Mechanism Distinction
- Refined Simulation Feedback Interactions (FIRE & STARFORGE)
- CO SLED Molecular Gas Tracers (ALMA)
- IR Starforming & Disk Signatures (JWST)

Thank You!

- R. C. Hickox & D. M. Alexander, *Obscured Active Galactic Nuclei* (2018) ARA&A Volume 56
- P. F. Hopkins et. al., *Stellar and Quasar Feedback in Concert: Effects on AGN Accretion, Obscuration and Outflows* (2016) MNRAS Volume 458 Issue 1

Resources on GitHub: [here](#)